Demo Client · DEMO-0006
11 November 2026 · v1.0

Field	Value
Document	Methodology of the automated monthly management report
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Engagement	DEMO-0006 · Demo Client
Report	Monthly Management Report, June 2024
Prepared by	ProQuant Analytics · info@proquantanalytics.com
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1. Purpose

The monthly management report is produced by a controlled process instead of by hand: the same checks, the same definitions and the same layout every month, with evidence that the numbers are right and a clear decision when they are not. The process prepares documents and email drafts; people review and send them.

2. Inputs

Data set	Size	Fingerprint (SHA-256)
sales	17,430 rows	`a358e6cf8a72120f`
feed_log	18 rows	`cf55ff0967b802ce`
targets	89 rows	`0b6a07ec1a0041c5`
manual_report	48 rows	`15134be39a0a9242`
recipients	9 rows	`738a39651009236f`

Each input is fingerprinted with SHA-256 on ingest and recorded in the run audit trail.

3. Feed controls and the release gate

Ten controls run on every month: the feed arrived; arrived on time; the lines match the sender's control total (after duplicates are removed); the volume is in line with the median of the previous months; the last order is close to the end of the month; the share of duplicate lines; the share of lines in the exceptions queue; the share of lines without a product line; a budget for every region; net revenue in line with the previous six months (robust deviation from the median, with the spread never below 10% of the median). An eleventh control compares the inputs of a month with the ledger (restatement).

Each control is PASS, WARN or FAIL against thresholds agreed at kickoff and stored in the configuration. A FAIL on a **blocking** control holds the report: it is stamped DRAFT and its emails are drafted as held. Anything else that is not a PASS is released with notes, which the report lists. The same controls run for every month received, so the history shows what the process would have done in the past.

4. Cleaning

Spelling of regions is harmonised through the client's list and alias table. Duplicate lines (same line identifier) are removed, the first one kept, and counted per month. A line goes to the **exceptions queue**, and is excluded from every figure, when its date is invalid or outside the month of its file, its region is unknown, its units are not positive, its gross amount is outside the business limits, or a discount, return or cost is negative. Values are never altered or guessed. A blank product line is labelled and kept. Every action is counted in the cleaning log.

5. KPI engine

A KPI is a formula over base measures (gross sales, discounts, returns, cost of goods, units, lines, orders, customers, new customers) and over KPIs defined above it. Formulas are checked against a whitelist before use. For each period the measures are computed from the lines of the slice (total, region, product line; month and year to date) and the formulas applied to them, so a ratio is never an average of ratios. **Orders** are distinct order numbers, not lines. A **new customer** is a customer whose first valid line in the data falls in the period.

6. Comparisons

For the report month: against the prior month, the same month of the previous year and the budget; for the year to date: against the same period of the previous year and the budget. Differences are relative for amounts and in percentage points for ratios. A status (better, worse, within tolerance) uses the budget if it is loaded, otherwise the previous year, otherwise the prior month. The budget of a scope is shown only when every region it contains has a target for every month it covers; a partial budget would understate the total.

7. Commentary

Sentences are assembled by rules from the variance table: headline, drivers (regions with the largest contribution to the change), regions more than the tolerance below budget, the product line with the largest change, margin and discount rate, returns against their six-month average, new customers, year to date, values far from the recent months, and the data notes. When a number needed by a sentence is missing, the sentence says so. The analyst reads the draft and may add notes (`review/commentary_notes.yaml`).

8. Proof against the manual report (parity)

Every figure of the manual report is compared with the automation's value for the same month, scope and KPI, with a relative tolerance for amounts and an absolute tolerance in points for ratios. A difference outside the tolerance is tested against the usual alternative definitions (order lines instead of orders; returns or discounts not deducted). If one reproduces the manual figure the difference is **explained** and needs a definition decision signed off by the client; otherwise it is **unexplained** and the client re-checks the manual figure. The verdict is Accepted, Accepted after the definitions are signed off, or Open.

9. Ledger and restatements

Each run stores, for every month, a fingerprint of its input lines and targets and a fingerprint of its KPI values, together with the net revenue and the release decision. If a month that was reported before now has different inputs, the control Restatement warns and the report states the change in net revenue.

10. Distribution

Recipients are matched to documents by delivery type: `pack` (summary and full report), `summary`, or `bulletin` (one page for the recipient's region). A draft email (`.eml`) with the right attachments is created for each recipient, with a ledger of recipients, subjects and attachment fingerprints. A held report produces drafts marked held. The process never sends email.

11. Business case

The hours of the manual process come from the client's own estimate per task; the automated hours are the review time the client agrees to spend plus the measured run time. The saving, the yearly value and the payback on the project fee follow from these assumptions, which the client can change.

12. Operations

The process runs from a wrapper script on a schedule of several days (the feed may be late). Exit codes tell the scheduler what happened: 0 released, 2 held, 3 feed not received yet, 1 failure. The operations runbook describes the daily routine, what each control means and how to recover.

13. Reproducibility

Inputs, configuration and code version are recorded in `audit.json`. Re-running the same inputs gives the same KPI fingerprints. Every table in the deliverables is a CSV in the run folder.

14. Limitations

The limits of this work are set out in the report. No deviations from the standard methodology.